

ICE-5117

Advance Artificial Intelligence



Lecture - 4

Natural Language Processing

Course Teacher

Prof. Dr. Md. Ashikur Rahman Khan

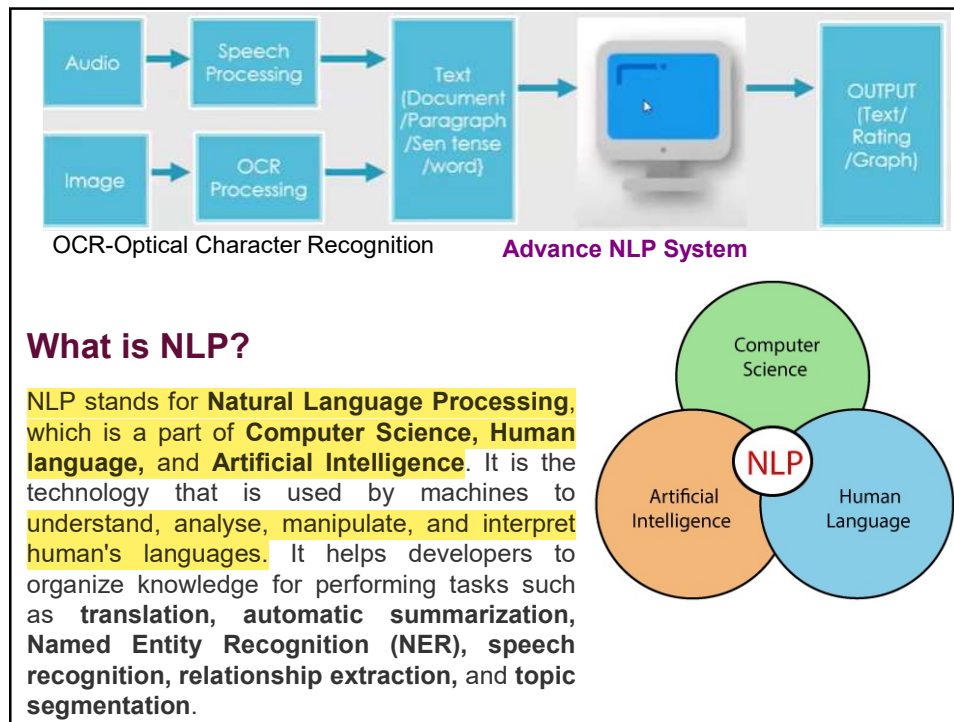
Natural Language Processing (NLP) refers to the branch of Artificial Intelligence that allows machines to understand human language

NLP is a fascinating field that combines linguistics, computer science, and artificial intelligence to enable computers to understand, interpret, and generate human language. Or

In other words, teaching computers how to understand and generate human language



General NLP System



Advantages of NLP

- NLP helps users to ask questions about any subject and get a direct response within seconds.
- NLP offers exact answers to the question means it does not offer unnecessary and unwanted information.
- NLP helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases.

Disadvantages of NLP

A list of disadvantages of NLP is given below:

- NLP may not show context.
- NLP is unpredictable
- NLP may require more keystrokes.
- NLP is unable to adapt to the new domain, and it has a limited function that's why NLP is built for a single and specific task only.

Components of NLP

There are the following two components of NLP -

- (U)nderstanding and (G)eneration



1. Natural Language Understanding (NLU)

Natural Language Understanding (NLU) helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.

NLU mainly used in Business applications to understand the customer's problem in both spoken and written language.

NLU involves the following tasks -

It is used to map the given input into useful representation.

It is used to analyze different aspects of the language.

2. Natural Language Generation (NLG)

Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.

Difference between NLU and NLG

NLU	NLG
NLU is the process of reading and interpreting language.	NLG is the process of writing or generating language.
It produces non-linguistic outputs from natural language inputs.	It produces constructing natural language outputs from non-linguistic inputs.

Examples of NLP (Understanding and Processing)

1.Sentiment Analysis (Social Media/Review Monitoring): NLP algorithms analyze customer reviews, social media posts, or survey responses to determine the emotional tone behind the text (positive, negative, or neutral). This allows companies to quickly gauge customer satisfaction without reading every comment.

2.Machine Translation (Google Translate): Modern translation tools use deep learning NLP to go beyond word-for-word translation, analyzing the context and grammatical structure of a sentence to provide accurate, fluent translations between languages.

Examples of NLG (Generating and Creating)

1.Conversational Chatbots/Virtual Assistants (e.g., ChatGPT, Siri): These systems use NLG to formulate human-like spoken or written responses in real-time, based on user input, enabling natural conversations.

2.Automated Financial/Data Reporting (e.g., Narrative Science): NLG systems can take structured data, such as a spreadsheet of daily sales figures or a company's financial report, and automatically transform that data into a written narrative report or summary.

- **Speech Recognition**
- **Sentiment Analysis (bad or good, or to analysis twitt)**
- **Machine Translation (exm simple google translation) which make easy the communication**
- **Chat Boot (Human and Machine communication)**
- **Spell checking**

Applications of NLP

There are the following applications of NLP -

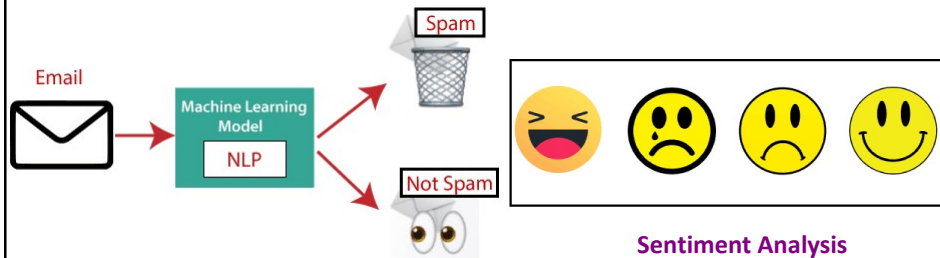
1. Question Answering

Question Answering focuses on building systems that automatically answer the questions asked by humans in a natural language.



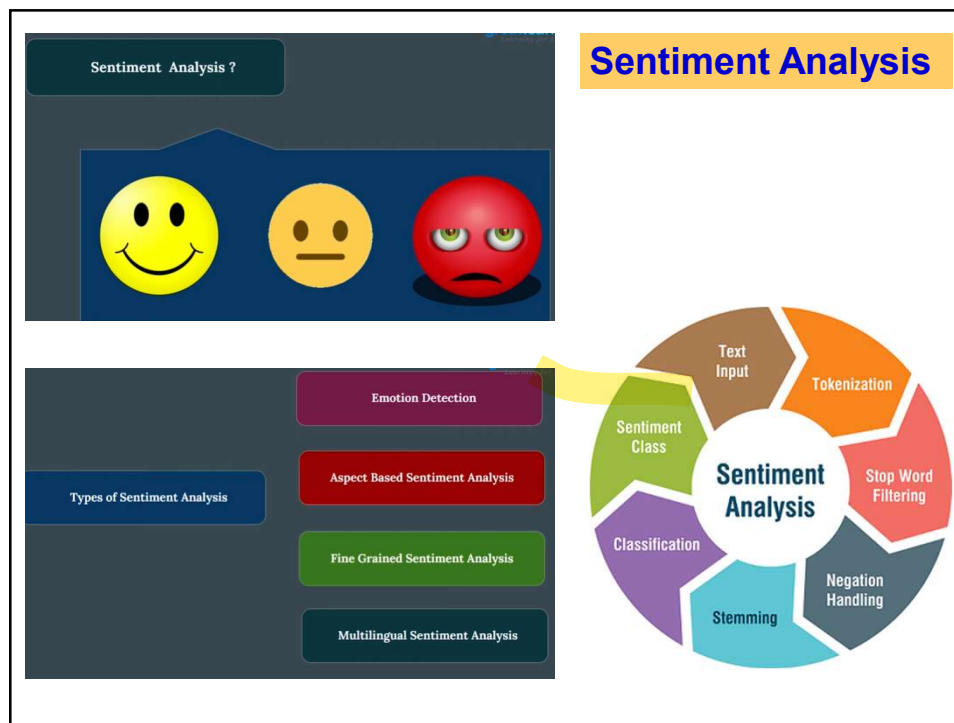
2. Spam Detection

Spam detection is used to detect unwanted e-mails getting to a user's inbox.



3. Sentiment Analysis

Sentiment Analysis is also known as **opinion mining**. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or neutral), identify the mood of the context (happy, sad, angry, etc.)



1. Emotion Detection

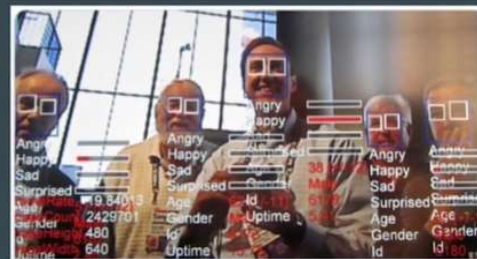
Emotion Detection



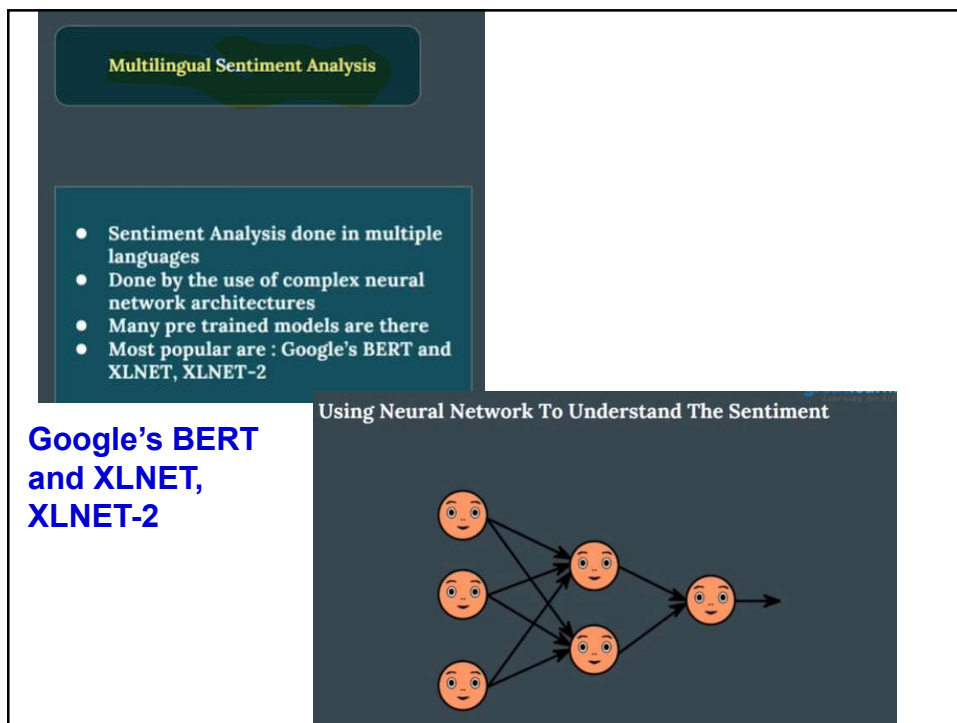
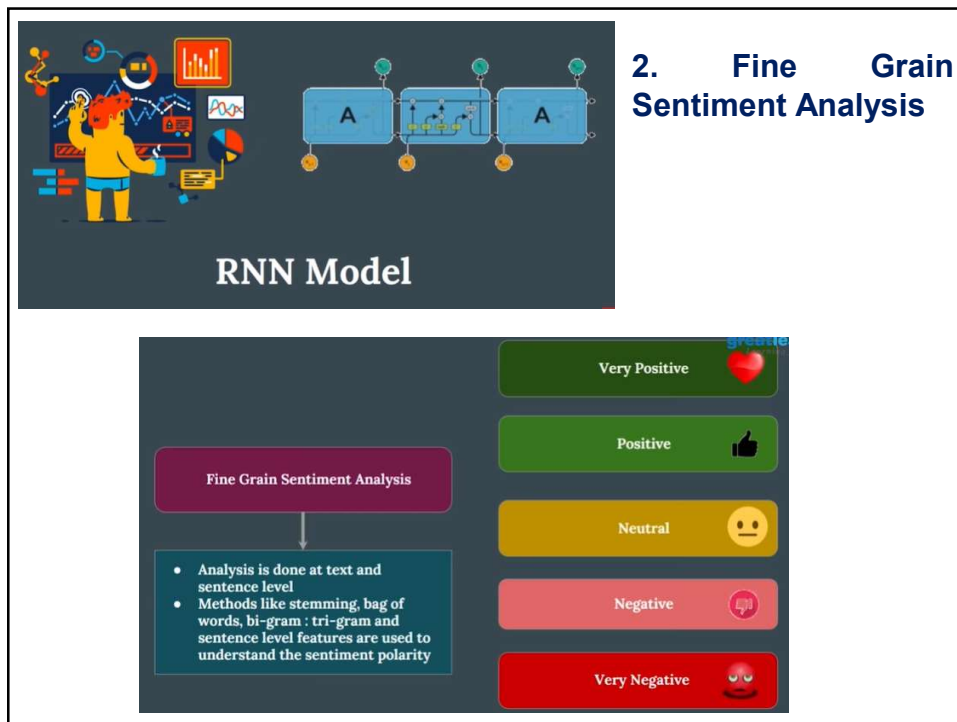
- **Emotion detection is the process of identifying human emotion.**
- People vary widely in their accuracy at recognizing the emotions of others.
- Use of technology to help people with emotion recognition is a relatively nascent research area.
- works best if it uses multiple **modalities** in context. Work has been conducted on automating the recognition of **facial expressions** from video, spoken expressions from audio, written expressions from text.

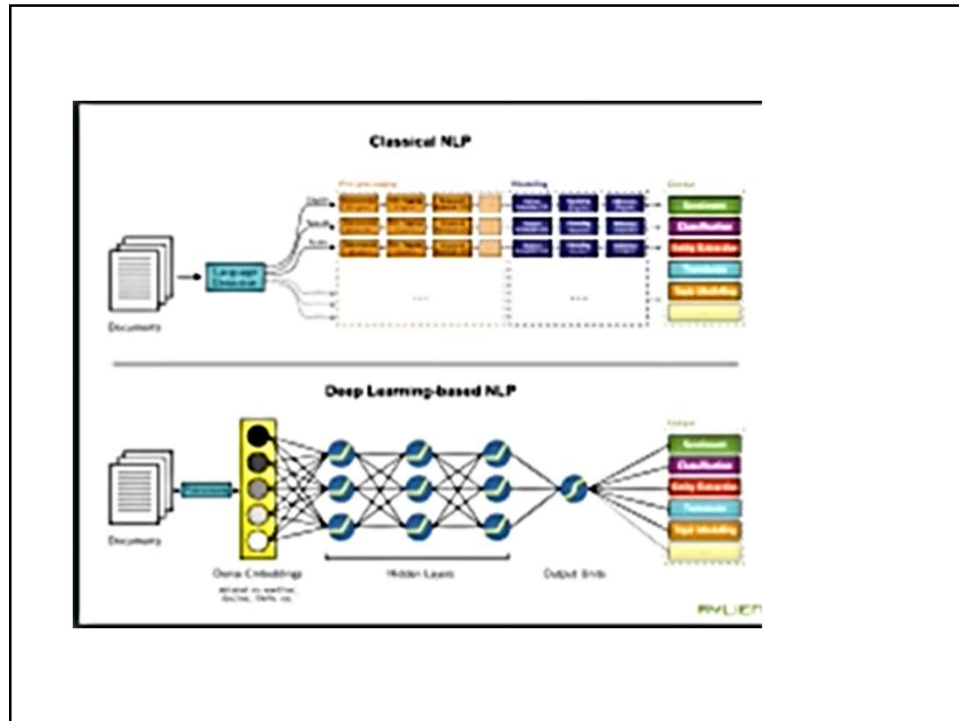


1. Emotion Detection



Algorithms used:
Naive Bayes, SVM, Neural Network, BERT



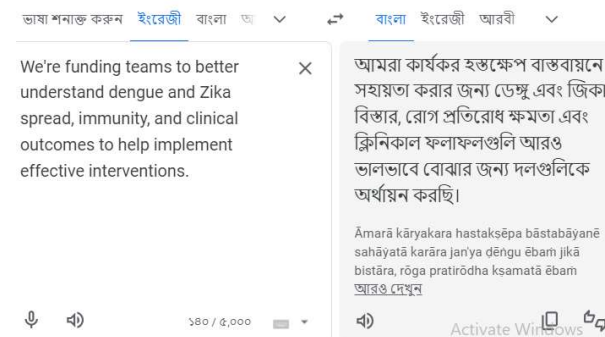


Feature	Emotion Detection	Aspect Based Sentiment	Fine Grained Sentiment	Multilingual Sentiment
Main goal	Detect emotion	Detect opinion about features	Detect intensity level	Detect sentiment in multiple languages
Output type	Joy, anger, sadness	sentiment for each aspect	rating scale (1–5)	positive/negative in many languages
Example	I feel sad	camera good, battery bad	very good, good, bad	Bangla, English, Arabic
Complexity	Medium	High	Medium	High
Application	psychology, chatbot	product review	rating analysis	global social media
Model type	classification	multi-label classification	regression/classification	multilingual NLP

4. Machine Translation

Machine translation is used to translate text or speech from one natural language to another natural language.

Example: Google Translator



5. Spelling correction

Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.

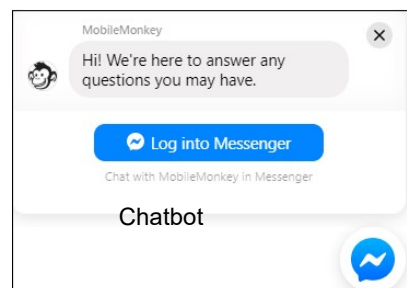


6. Speech Recognition

Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.

7. Chatbot

Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.



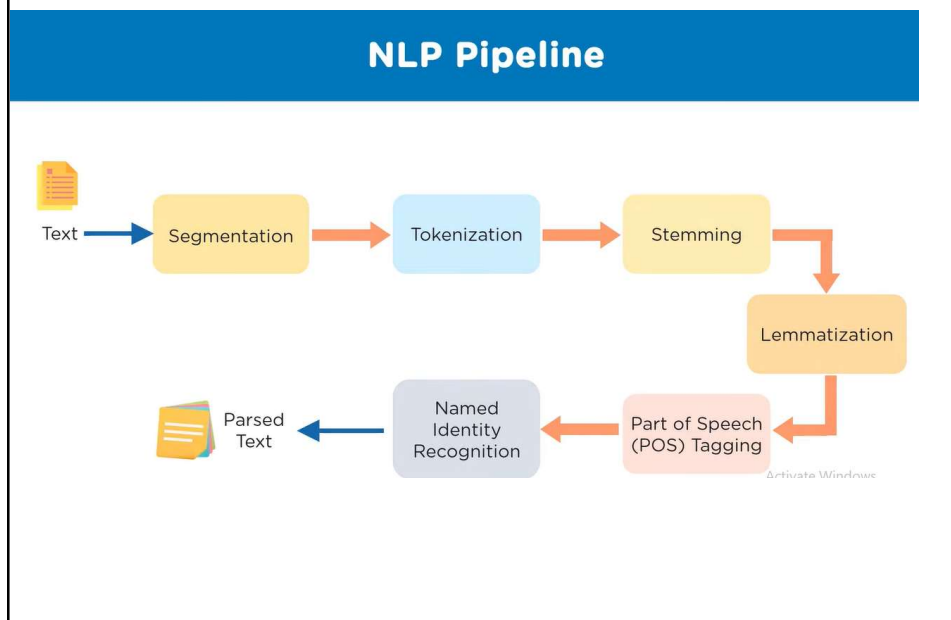
8. Information extraction

Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

9. Natural Language Understanding (NLU)

It converts a large set of text into more formal representations such as first-order logic structures that are easier for the computer programs to manipulate notations of the natural language processing.

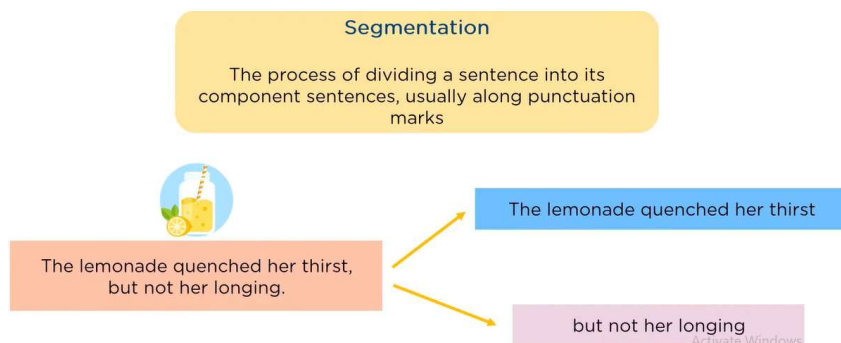
How to build an NLP pipeline



There are the following steps to build an NLP pipeline –

Step1: Sentence Segmentation

Sentence Segment is the first step for building the NLP pipeline. It's the process of dividing a sentence into its component sentences, usually along punctuation marks. It breaks the paragraph into separate sentences.



Example 2: Consider the following paragraph -

Independence Day is one of the important festivals for every Bangladeshi citizen. It is celebrated on the 26th of March each year ever since Bangladesh got independence from the Pakistan rule. The day celebrates independence in the true sense.

Sentence Segment produces the following result:

"Independence Day is one of the important festivals for every Bangladeshi citizen."

"It is celebrated on the 26th of March each year ever since Bangladesh got independence from the Pakistan rule."

"This day celebrates independence in the true sense."

The first step in the pipeline is to break the text apart into separate sentences. That gives us this:

1. "Copenhagen is the capital and most populous city of Denmark and the coastal islands."

2. "The coastal islands of Scandinavia, Region in Europe Great Britain, Copenhagen has been a major settlement for two millennia 50 thousand two."

Home Work:

Copenhagen is the capital and most populous city of Denmark and capital sits on the coastal islands of Zealand and Amager. It's linked to Malmo in southern Sweden by the Oresund Bridge. Indre By, the city's historic centre, contains Frederiksstaden, an 18th-century rococo district, home to the royal family's Amalienborg Palace. Nearby is Christiansborg Palace and the Renaissance-era Rosenborg Castle, surrounded by gardens and home to the crown jewels.

All full stops are not the closing of sentence. For example Dr. Ph.D etc.

So, there are two classes

one is end of sentence

not end of sentence

is called classification

Sentence Segmentation

The problem of deciding where the sentences begin and end.

Challenges Involved

- While '!', '?' are quite unambiguous
- Period "." is quite ambiguous and can be used additionally for
 - Abbreviations (Dr., Mr., m.p.h.)
 - Numbers (2.4%, 4.3)

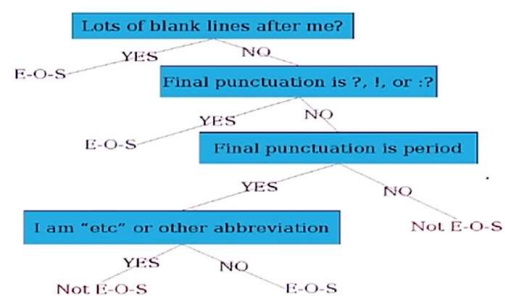
Approach: build a binary classifier

For each "."

- Decides EndOfSentence/NotEndOfSentence
- Classifiers can be: hand-written rules, regular expressions, or machine learning

- !, ? are relatively unambiguous
- Period "." is quite ambiguous
 - Sentence boundary
 - Abbreviations like Inc. or Dr.
 - Numbers like .02% or 4.3
- Build a binary classifier
 - Looks at a "."
 - Decides EndOfSentence/NotEndOfSentence
 - Classifiers: hand-written rules, regular expressions, or machine-learning

Decision Tree: Is this word the end-of-sentence (E-O-S)?



Other Important Features

- punctuation
- formatting
- fonts
- spacing
- capitalization
- case
- use of abbreviations, e.g., Dr., a.m.

Example

- If there is no space after a period, don't assume that there is a sentence boundary
- Case of word with ".": Upper, Lower, Cap, Number
- Case of word after ".": Upper, Lower, Cap, Number
- Numeric Features
 - ▶ Length of word with "."
 - ▶ Probability (word with "." occurs at end-of-sentence)
 - ▶ Probability (word after "." occurs at beginning-of-sentence)

Step2: Word Tokenization

The next step in our pipeline is to break this sentence into separate words or tokens. This is called tokenization.

Word Tokenizer is used to break the sentence into separate words or tokens.

Example 1:

"Copenhagen is the capital and most populous city of Denmark and the coastal islands"

Word Tokenizer generates the following result:

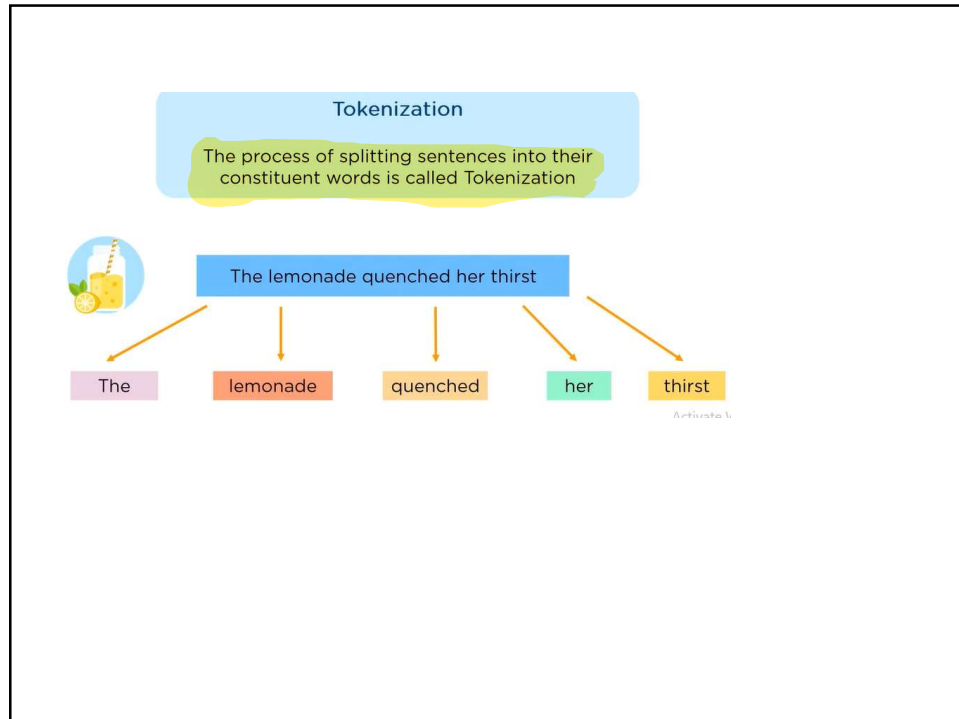
"Copenhagen", "is", "the", "capital", "and", "most", "populous", "city", "of", "Denmark", "and", "the", "coastal", "islands", "."

Example 2:

JavaTpoint offers Corporate Training, Summer Training, Online Training, and Winter Training.

Word Tokenizer generates the following result:

"JavaTpoint", "offers", "Corporate", "Training", "Summer", "Training", "Online", "Training", "and", "Winter", "Training", "."



Space-based tokenization

A very simple way to tokenize

- For languages that use space characters between words
- Arabic, Cyrillic, Greek, Latin, etc., based writing systems
- Segment off a token between instances of spaces

Issues in Tokenization

Can't just blindly remove punctuation:

- [m.p.h.](#), [Ph.D.](#), [AT&T](#), [cap'n](#)
- prices ([\\$45.55](#))
- dates ([01/02/06](#))
- URLs (<http://www.stanford.edu>)
- hashtags ([#nlproc](#))
- email addresses (someone@cs.colorado.edu)

Clitic: a word that doesn't stand on its own

- "are" in [we're](#), French "je" in [j'ai](#), "le" in [l'honneur](#)

Tokenization in languages without spaces

Many languages (Chinese, Japanese, Thai) don't use spaces to separate words!

How do we decide where the token boundaries should be?

Word tokenization in Chinese

Chinese words are composed of characters called "hanzi" (or sometimes just "zi")

Each one represents a meaning unit called a morpheme.

Each word has on average 2.4 of them.

But deciding what counts as a word is complex and not agreed upon.

Word tokenization / segmentation

So in Chinese it's common to just treat each character (zi) as a token.

- So the **segmentation** step is very simple

In other languages (like Thai and Japanese), more complex word segmentation is required.

- The standard algorithms are neural sequence models trained by supervised machine learning.

Step3: Stemming

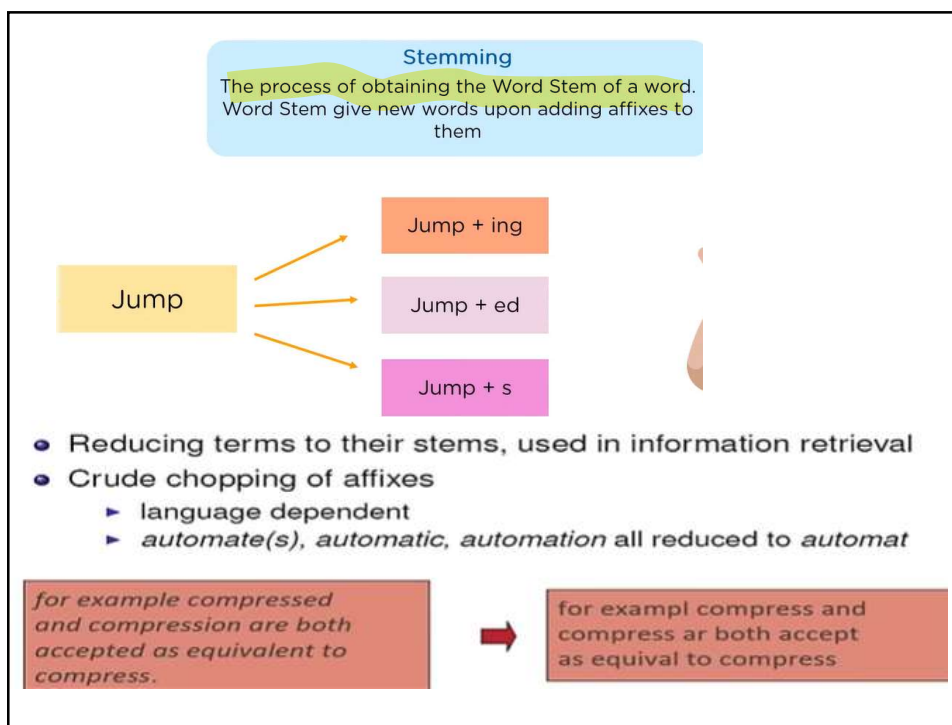
Stemming is used to normalize words into its base form or root form.

For example, celebrates, celebrated and celebrating, all these words are originated with a single root word "celebrate."

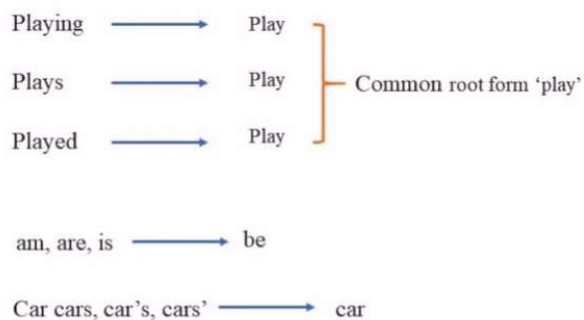
The big problem with stemming is that sometimes it produces the root word which may not have any meaning.

For Example, intelligence, intelligent, and intelligently, all these words are originated with a single root word "intelligen."

In English, the word "intelligen" do not have any meaning.



Stemming vs Lemmatization



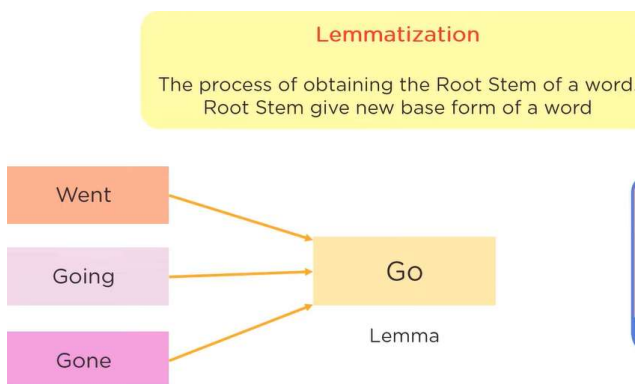
Using above mapping a sentence could be normalized as follows:

the boy's cars are different colors → the boy car be differ color

Step 4: Lemmatization

Lemmatization is quite similar to the Stemming. It is used to group different inflected forms of the word, called Lemma. The main difference between Stemming and lemmatization is that it produces the root word, which has a meaning.

For example: In lemmatization, the words intelligence, intelligent, and intelligently has a root word intelligent, which has a meaning.



- Reduce inflections or variant forms to base form:
 - ▶ am, are, is → be
 - ▶ car, cars, car's, cars' → car
- Have to find the correct dictionary headword form

am, are, is → be

Car cars, car's, cars' → car

Using above mapping a sentence could be normalized as follows:

the boy's cars are different colors → the boy car be differ color

Step 5: Identifying Stop Words

In English, there are a lot of words that appear very frequently like "is", "and", "the", and "a". NLP pipelines will flag these words as stop words. **Stop words** might be filtered out before doing any statistical analysis.

Example: He is a good boy.

Step 6: Dependency Parsing

Dependency Parsing is used to find that how all the words in the sentence are related to each other.

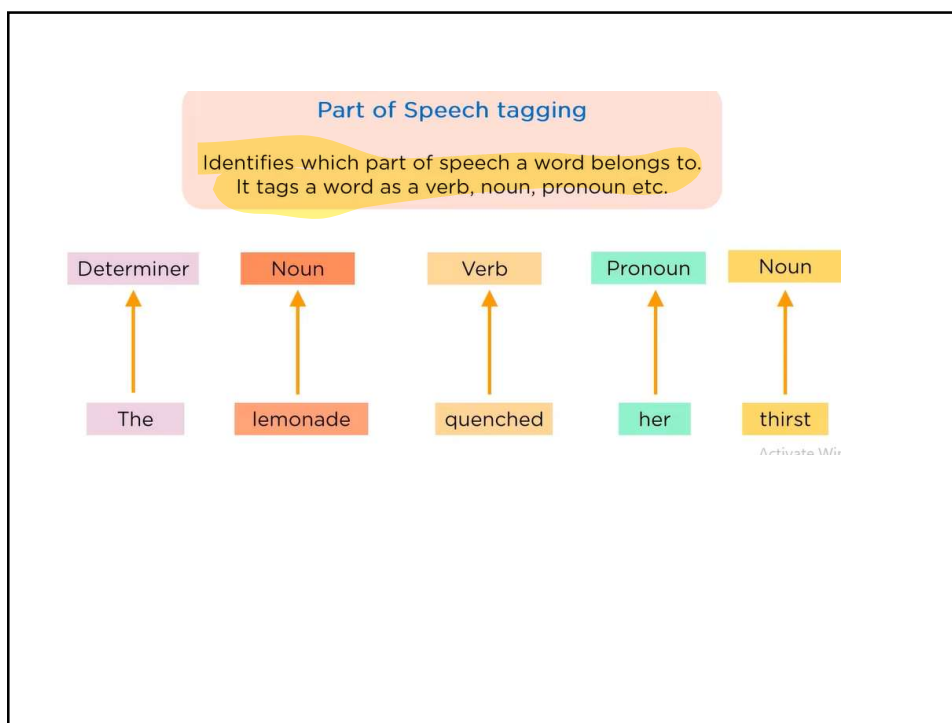
Step 7: POS (Parts of Speech) tags

POS stands for parts of speech, which includes Noun, verb, adverb, and Adjective. It indicates that how a word functions with its meaning as well as grammatically within the sentences. A word has one or more parts of speech based on the context in which it is used.

Example: "Google" something on the Internet.

In the above example, Google is used as a verb, although it is a proper noun.

Copenhagen	is	the	Capital	and	most	populous
Proper Noun	Verb	Determiner	Noun	Conjunction	Adverb	Adjective



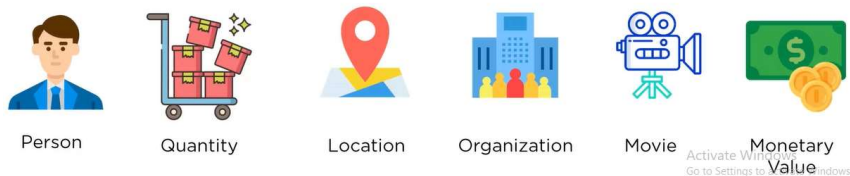
Step 8: Named Entity Recognition (NER)

Named Entity Recognition (NER) is the process of detecting the named entity such as person name, movie name, organization name, or location.

Example: Steve Jobs introduced iPhone at the Macworld Conference in San Francisco, California.

Named Entity Recognition

Classifying the words into subcategories.
The subcategories are :



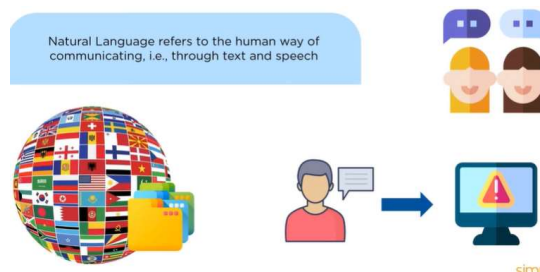
Step 9: Chunking

Chunking is used to collect the individual piece of information and grouping them into bigger pieces of sentences.

Natural Language refers to the human way of communicating, i.e., through text and speech.

There are so many languages in the world.

However, the machine cannot understand word (text, speech). They cannot understand human's languages such as English, French etc. It can understand only binary 0 and 1.



Why NLP is difficult? Or Challenges of NLP

NLP is difficult because Ambiguity and Uncertainty exist in the language.

Ambiguity

There are the following three ambiguity -

Lexical Ambiguity

Lexical Ambiguity exists in the presence of two or more possible meanings of the sentence within a single word.

Example:

Manya is looking for a **match**.

In the above example, the word match refers to that either Manya is looking for a partner or Manya is looking for a match. (Cricket or other match)

Why NLP is difficult? Or Challenges of NLP

Syntactic Ambiguity

Syntactic Ambiguity exists in the presence of two or more possible meanings within the sentence.

Example:

I saw the girl with the binocular.

In the above example, did I have the binoculars? Or did the girl have the binoculars?

Referential Ambiguity

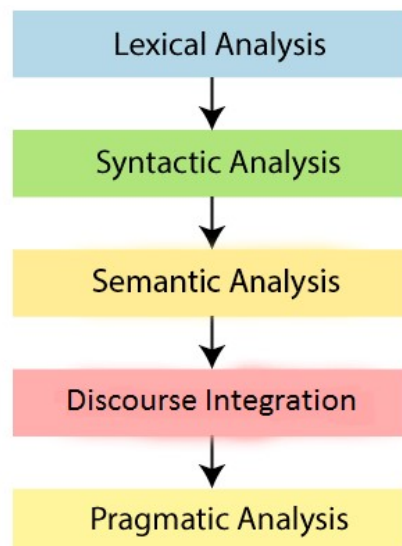
Referential Ambiguity exists when you are referring to something using the pronoun.

Example: Kiran went to Sunita. She said, "I am hungry."

In the above sentence, you do not know that who is hungry, either Kiran or Sunita.

Phases of NLP

There are the following five phases of NLP:



1. Lexical Analysis and Morphological

The first phase of NLP is word structure analysis, which is referred to as lexical or morphological analysis. This phase scans the source code (input text etc.) as a stream of characters and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

A lexicon is defined as a collection of words and phrases in a given language, with the analysis of this collection being the process of splitting the lexicon into components, based on what the user sets as parameters – paragraphs, phrases, words, or characters.

There are 2 types of morphemes:

Free morphemes function independently as words (like “cow” and “house”).

Bound morphemes make up larger words. The word “unimaginable” contains the morphemes “un-” (a bound morpheme signifying a negative context), “imagine” (the free morpheme root of the whole word), and “-able” (a bound morpheme denoting the root morpheme’s ability to end).

For instance, when performing a lexical analysis on the previous paragraph, the analysis isolates and divides the first sentence into lexeme phrases, like “the understandable vocabulary that makes up a language.” This analysis further divides the phrase into word lexemes, like “vocabulary” and “language,” categorizing both as noun POS. Then, the analysis derives free morphemes, like “words,” “vocabulary,” and “understand-,” and bound morphemes, like “-able.”

2. Syntactic Analysis (Parsing)

Syntax Analysis is the second phase of natural language processing. Syntactic Analysis or parsing is the process of checking grammar, word arrangements, and shows the relationship among the words and whether those make sense.

The process involved examination of all words and phrases in a sentence, and the structures between them.

Syntax describes how a language's words and phrases arrange to form sentences.

Syntactic analysis checks word arrangements for proper grammar.

For instance, the sentence "Dave paper" passes a syntactic analysis check because it's grammatically correct.

Conversely, a syntactic analysis categorizes a sentence like "Dave do jumps" as syntactically incorrect.

Example: Agra goes to the Poonam

In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis

Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal (accurate) meaning of words, phrases, and sentences.

Semantics describe the meaning of words, phrases, sentences, and paragraphs. Semantic analysis attempts to understand the literal meaning of individual language selections, not syntactic correctness. However, a semantic analysis doesn't check language data before and after a selection to clarify its meaning.

For instance, "Manhattan calls out to Dave" passes a syntactic analysis because it's a grammatically correct sentence. However, it fails a semantic analysis. Because Manhattan is a place (and can't literally call out to people), the sentence's meaning doesn't make sense.

Jhon eat mango. Right but it is as follow

Mango eats jhon. Wrong

4. Discourse Integration

Discourse Integration depends upon the sentences that precedes it and also invokes the meaning of the sentences that follow it.

Discourse describes communication between 2 or more individuals. Discourse integration analyzes prior words and sentences to understand the meaning of ambiguous language.

For instance, if one sentence reads, "Manhattan speaks to all its people," and the following sentence reads, "It calls out to Dave," discourse integration checks the first sentence for context to understand that "It" in the latter sentence refers to Manhattan.

5. Pragmatic Analysis

Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.

For Example: "Open the door" is interpreted as a request instead of an order.

Pragmatism describes the interpretation of language's intended meaning. Pragmatic analysis attempts to derive the intended—not literal—meaning of language.

For instance, a pragmatic analysis can uncover the intended meaning of "Manhattan speaks to all its people." Methods like neural networks assess the context to understand that the sentence isn't literal, and most people won't interpret it as such. A pragmatic analysis deduces that this sentence is a metaphor for how people emotionally connect with places.

Handling Hyphenation

Hyphens can be

End-of-Line Hyphen

Used for splitting whole words into part for text justification.

This paper describes MIMIC, an adaptive mixed initiative spoken dialogue system that provides movie show-time information.

Lexical Hyphen

Certain prefixes are often written hyphenated, e.g. co-, pre-, meta-, multi-, etc.

Sententially Determined Hyphenation

Mainly to prevent incorrect parsing of the phrase. Some possible usages:

- Noun modified by an 'ed'-verb: *case-based, hand-delivered*
- Entire expression as a modifier in a noun group: *three-to-five-year direct marketing plan*

Longest Words of Distinct Languages

Max ▾	Language (non scientific) ♦
431	Sanskrit (Longest)
173	Greek
136	Afrikaans
85	Māori
79	German
74	Turkish
64	Icelandic
56	Hungarian
54	Spanish
49	Dutch
46	Malay
45	English

44	Romanian
42	Georgian
41	Czech
39	Bulgarian
39	Lithuanian
36	Kazakh
33	Norwegian
32	Tagalog
32	Polish
30	Serbian
30	Montenegrin
30	Italian
30	Croatian

Reference Books:

- Speech and Language Processing: An Introduction to Natural Language Processing, Speech Recognition, and Computational Linguistics. 2nd Edition. Prentice-Hall. By Daniel Jurafsky and James H. Martin.
- Foundations of Statistical Natural Language Processing. MIT Press. Christopher D. Manning and Hinrich Schtza

